

ROLL NO:250701573

EXPERIMENT NO: 17 RANDOM SAMPLING AND SAMPLING DISTRIBUTION

PROGRAM:

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Step 1: Generate a population
```

```
population_mean = 65
```

```
population_std = 12
```

```
population_size = 100000
```

```
population = np.random.normal(population_mean,  
                               population_std,  
                               population_size)
```

```
# Step 2: Random sampling
```

```
sample_sizes = [25, 50, 80]
```

```
num_samples = 800
```

```
sample_means = {}
```

```
for size in sample_sizes:
```

```
    sample_means[size] = []
```

```
    for _ in range(num_samples):
```

```
        sample = np.random.choice(population,
```

```

        size=size,
        replace=False)

    sample_means[size].append(np.mean(sample))

# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):

    plt.subplot(len(sample_sizes), 1, i+1)

    plt.hist(sample_means[size],
             bins=30,
             alpha=0.7,
             label=f'Sample Size {size}')

    plt.axvline(np.mean(population),
               color='red',
               linestyle='dashed',
               linewidth=1.5,
               label='Population Mean')

    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

```

```
plt.tight_layout()
```

```
plt.show()
```

OUTPUT:

